Ryan R. Hu

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Education

California Institute of Technology (Caltech) Pasadena, CA
B.S. Computer Science - *GPA: 4.2/4.0* *Graduation Date: Jun 2027*

Relevant Coursework: Computer Programming, Programming Methods, Software Design, Deep Learning, Learning Systems, Machine Learning & Data Mining, Algorithms, Computing Systems, Diffusion Models, Differential Equations, Mathematical Foundations of Computer Science (Discrete Mathematics)

Teaching Assistant: Decidability and Tractability (Jan 2025 - Mar 2025), Machine Learning & Data Mining (Jan 2026 - Mar 2026)

Technical Skills

Languages: Python, C, C++, Java, JavaScript/TypeScript, HTML/CSS, SQL, MATLAB, LaTeX
Developer Tools: Git, VS Code, AWS, Github, Linux, Docker, Kubernetes, Figma
Frameworks/Libraries: React.js, Node.js, Pandas, OpenCV, PyTorch, Keras, TensorFlow, Scikit, Tailwind, Flask, Plotly, PySpark
Skills: CI/CD, Machine Learning, Full Stack, Data Analysis, Debug Logging, Containerization, Unit Testing

Experience

Software Development Engineer Intern June 2026 – Sep 2026
Amazon - Business Payments Team *Seattle, WA*
- Incoming Summer 2026

Machine Learning Researcher April 2026 – Present
Caltech - Wierman Lab *Pasadena, CA*
- Researching diffusion-based approaches for constrained optimal power flow in energy system contexts.

Machine Learning Researcher Feb 2025 – Present
Caltech - Alvarez Lab *Pasadena, CA*
- Building custom autoencoder architecture to detect anomalous player behavior in Activision's Call of Duty (COD) games, using SQL and Python to clean and process 3M+ raw gameplay logs, identifying 1000+ anomalous players within a 10-day window through reconstruction error.
- Designing a verification pipeline integrating UMAP + HDBSCAN clustering with temporal segmentation, identifying distinct types of anomalous behaviors and analyzing their impact on player engagement in 20,000+ players.

Machine Learning Intern Jun 2025 – Sep 2025
Tracevision - Math Team *Los Angeles, CA*
- Automated data preprocessing pipelines and scripts to extract and construct 6000+ shot sequences from 80+ hours of raw basketball game footage centered around the basket, optimizing video format and quality for annotation teams through frame-level segmentation and noise reduction.
- Designed and deployed production-grade hybrid CNN + Transformer shot detection model, leveraging MobileNetV2 as the CNN backbone, achieving 95% accuracy on made vs. missed shot classification from annotated video clips.

Software Engineer Intern Apr 2025 – Sep 2025
Tracevision - Math Team *Los Angeles, CA*
- Improved ptools, a full-stack internal tool for object tracking and annotation review, by optimizing components to boost efficiency and enhance usability for 100+ users - cutting detection load latency from 1.2s to 0.7s ($\sim$ 40% faster). Built with Python, MySQL, HTML/CSS, and JavaScript.
- Developed geospatial API deployment pipelines using a combination of custom scripts and AWS infrastructure (EC2, S3, Lambda), streamlining integration workflows for 30+ facilities and 100+ cameras in security and retail sectors.

Undergraduate Researcher Apr 2024 – Dec 2024
Caltech - Alvarez Lab *Pasadena, CA*
- Analyzed HIV patient datasets collected by the Multicenter AIDS Cohort Study (MACS) by applying causal ML and gradient boosting trees on 800+ patients, discovering significant epigenetic markers correlated with accelerated aging - currently authoring manuscripts on findings.
- Built data preprocessing pipeline merging epigenetic datasets with web-scraped environmental satellite data, performing data cleaning, imputation, and feature transformation for machine learning tasks downstream.

Undergraduate Researcher Jun 2023 – Sep 2023
Caltech - GALCIT Lab *Pasadena, CA*
- Developed a visual anemometry framework in MATLAB combining YOLOv3 object detection (95.4% precision) with optical flow mapping techniques to estimate wind speeds within ±5 m/s error.
- Conducted a quantitative analysis linking canopy motion variance to wind standard deviation across three different tree species, achieving $>$0.9 correlation in field tests.

Selected Projects

TinyLM | *Python, FastAPI, React, PyTorch*
- Implemented a lightweight transformer-based language model ($\sim$22M total parameters) trained on TinyStories dataset. Includes custom BPE tokenization, dataset pipeline, training loop, and autoregressive text generation. Exposed inference through FastAPI backend for real-time prompting and querying.
- Designed a full-stack web interface with React connected to the FastAPI backend, enabling prompt-based interaction and live text generation.

Agentic Stock Trader | *Python, LangChain, GCP, Gemini API*
- Built a multi-agent stock trading framework where LLM-driven agents ingest 1,000+ financial news items, debate with each other, and output explainable decisions, achieving up to 2.34% monthly return for various individual stocks in backtests.
- Designed a recursive agent debate architecture with static personas, dynamic leaf agents, and hierarchical clustering, creating 50+ distinct agents across 3 debate layers and multiple decision rounds to systematically evaluate trades and reach agreement.

Endless Runner Video Game | *C, HTML/CSS, SDL*
- Developed Google Dino Run-inspired endless runner game with obstacles, powerups, and progressive difficulty scaling. Written entirely in C using standard libraries.
- Implemented 100% unit test coverage for the physics engine. Reduced collision computation by $\sim$40% overhead by restricting hitbox checks to active on-screen segments only.

Activities/Achievements

NCAA Division III Men's Basketball 2024, 2025
SCIAC All-Academic Team 2024, 2025
Hacktech Organizer - Caltech's Hackathon 2025, 2026
National Merit Finalist 2023